

VERITRANSIT

AI-powered physical verification for B2B commerce

A worker copilot that checks what physically arrived against the digital order — and writes the record itself.

Android + on-device AI

camera, NPU, works offline

Kotlin backend

reconciliation, audit chain

Ops agent

records, reports, alerts

Not a compliance tool. A faster pair of hands on the dock.



THE PREMISE

Every B2B transaction has two realities

One is structured, auditable and trusted by finance. The other is on a pallet, and gets confirmed by eye.

THE DIGITAL ORDER

Exact, structured, already in the system

- Purchase order — 10 units, SKU, rate, PO line
- Tax invoice — value, tax, payment terms
- Manifest / delivery note — what was declared

Machine-readable. Finance acts on it.

**THE
GAP**

THE PHYSICAL SHIPMENT

Counted by eye, written on paper

- Cartons that may be short, extra or swapped
- Labels that are missing, torn or unreadable
- Damage nobody logs until the customer calls

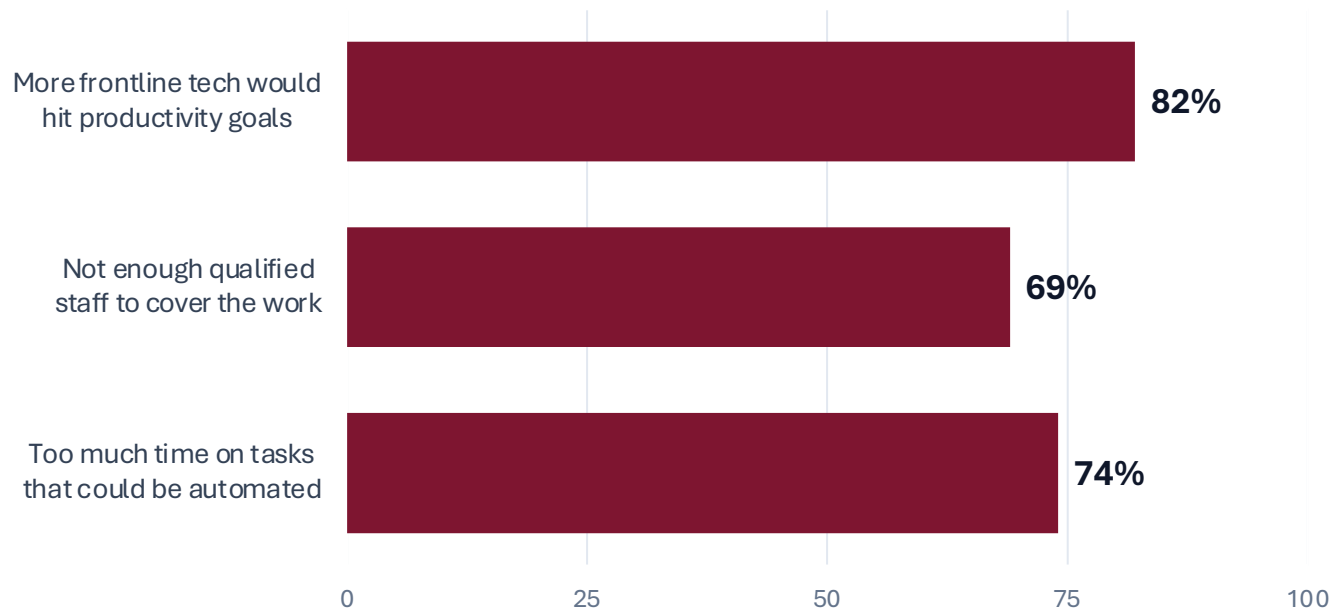
Opaque to every system downstream.

The mismatch between the two is discovered weeks later, in an invoice dispute — by people who never saw the cargo.

The frontline is already telling us this work should be automated

Warehouse associates and the leaders who staff them agree on the diagnosis.

What warehouse workers and leaders report



74%

of workers say they spend too much time on tasks a machine could do — while 69% of operations cannot find enough qualified staff to do them.

Where that time actually goes

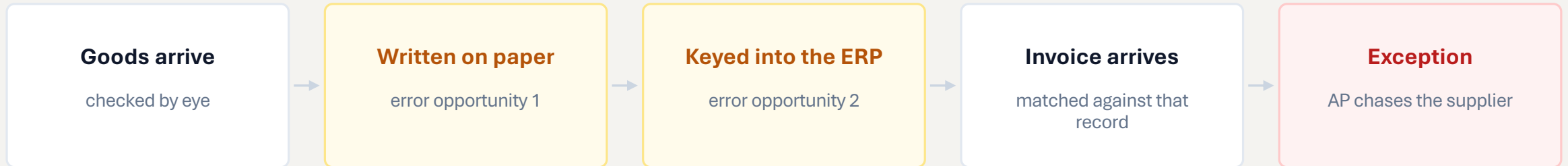
- **Counting and checking cargo**
by eye, carton by carton
- **Writing it on a tally sheet**
then keying it into the ERP
- **Chasing what does not add up**
days after the truck has gone

The demand is not for another system to update. It is for the checking itself to stop being manual.

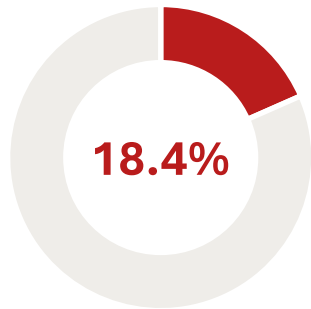
Source: Zebra Technologies, Warehousing Vision Study 2025 (1,700+ warehouse associates and decision-makers).

Manual receiving creates the errors that finance pays for later

Two hand-offs, two chances to be wrong — and the bill arrives in accounts payable.



Invoices that hit an exception



Nearly one invoice in five needs a human to reconcile it against what was actually received.

2

hand-offs between the dock and the ledger

The goods are recorded twice by hand before any system compares them to the order.

Weeks

until a short shipment surfaces

Typically at invoice matching or a payment dispute — long after the evidence is gone.

Sources: Ardent Partners, Accounts Payable Metrics that Matter 2025 (average invoice exception rate, 18.4%); Oracle, on paper-then-ERP receiving creating two separate opportunities for human error.

Three things just became true — and make this buildable

The demand, the hardware and the digital counterpart all arrived at the same time.

01

The market is buying frontline tech

82% of warehouse decision-makers say more technology in workers' hands is how they hit productivity targets. The budget line exists — it is labelled augmentation, not compliance.

82%

leaders backing frontline tech

02

The phone can now see

A mid-range handset reads several codes at once and runs a vision model on its own NPU. No rugged scanner to buy, no bandwidth at the dock, no photo leaving the site.

0

new hardware per worker

03

The digital side is machine-readable

POs, e-invoices and e-way bills are structured data. Once the physical side becomes structured too, comparing them is arithmetic — not paperwork.

1

missing half of the equation

Source: Zebra Technologies, Warehousing Vision Study 2025. Structural context: NITI Aayog on fragmented logistics, duplicate processes, manual handling and non-uniform documentation as efficiency constraints.

Today's stack checks documents. Nothing checks the goods.

Each layer proves something real — and each one assumes the layer below it was honest.

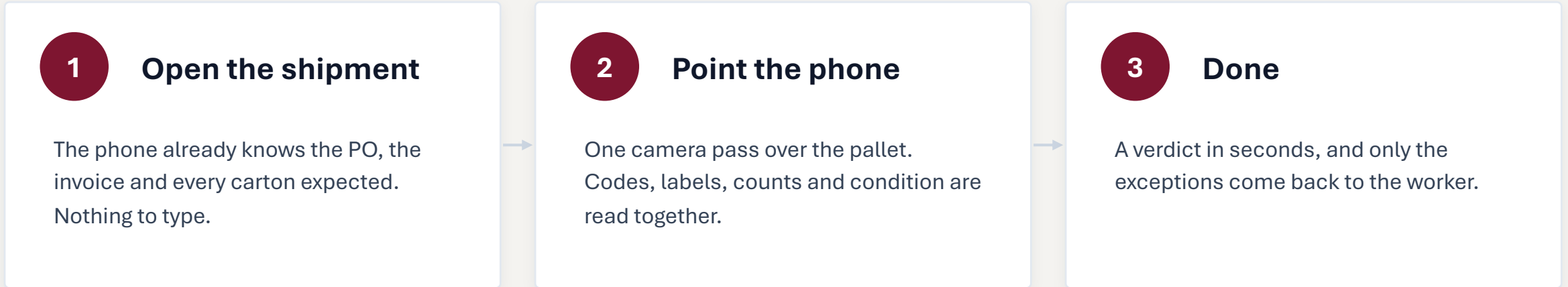
SYSTEM IN PLACE TODAY	WHAT IT PROVES	WHAT IT CANNOT SEE
WMS / ERP	What should arrive	Has no idea what did
Barcode scanner	That a code was scanned	Not that the code matches the contents
AP 3-way match	Paper against paper	Trusts a goods receipt typed by hand
VeriTransit	What is physically there	Turns the cargo itself into structured data, at the moment a human is already standing in front of it.

We are not replacing the WMS, the scanner or the ledger. We are supplying the one input all three currently guess at.

THE SOLUTION

Open the shipment. Point the phone. Done.

The worker's entire interaction. Everything else happens behind that.



The worker should not be filling in forms while the AI watches.

Every screen we add is a screen someone has to tap through at 6am on a loading dock. The AI exists to remove the boring work — the counting, the tallying, the typing — not to supervise a human who is still doing it.

We do not replace the worker. We make the worker dramatically faster.

Four things happen in that single camera pass

Three independent readings of the same pallet, then one comparison against the order.

READ

Scan every code at once

- Multiple QR and barcodes in one frame
- Identifies the packages that already carry codes
- No one-carton-at-a-time trigger pulling

READ

Read the digital order

- PO, invoice, manifest or delivery note
- Photographed and parsed on the device
- Gives the pass something to be right about

SEE

Look at the cargo

- Counts cartons and identifies products
- Reads printed labels; flags damage and re-taping
- Catches packages whose code is missing or unreadable

DECIDE

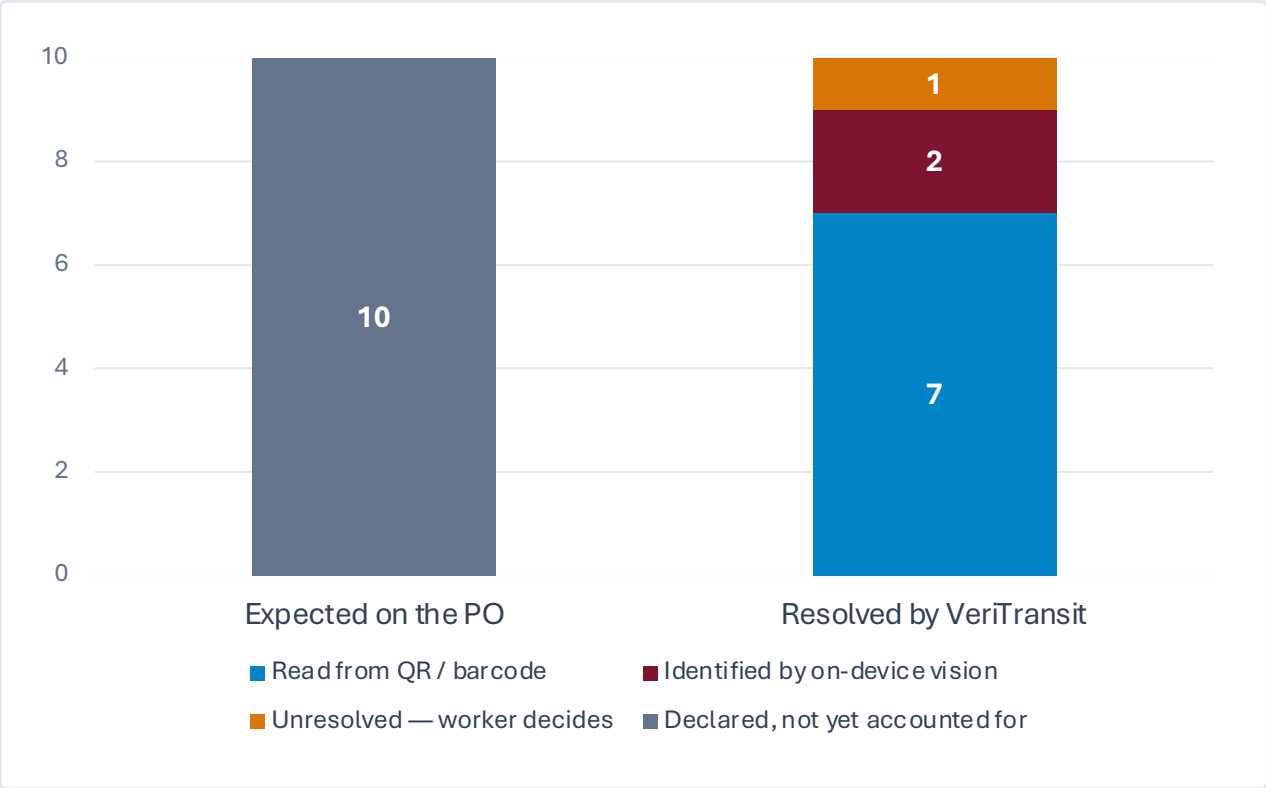
Reconcile the two sides

- Physical set vs declared set, line by line
- Every unit attributed to a source of truth
- Anything unattributed becomes an exception

All four run on the device. A dock with no signal still gets an answer — and no photo of a customer's cargo leaves the site.

Every unit is attributed — or it becomes an exception

A worked example: ten units on the PO, and what the pass resolves them to.



WHAT THE WORKER'S PHONE COMPUTES	
Expected	10
QR identified	7
Vision identified	2
Unknown	1

1 package needs a look

Nine units are closed out automatically. The worker is handed one carton and a reason — not a list of ten to re-check.

The output is not a score or a probability. It is an attribution for every unit, which is what a goods receipt legally needs.

Worked example on PO ZD-4471; the same arithmetic runs on any line count.

The worker sees three outcomes, and acts on one of them

Attention is the scarce resource on a dock. We spend it only where a human is actually needed.



MATCH

Nothing to do

Physical set equals the declared set. The goods receipt is already written by the time the worker looks up.

the overwhelming majority of cartons



CHECK THIS ONE

One carton, one question

A code would not read, a count is ambiguous, packaging looks disturbed. The phone says which carton and why.

the only thing the worker handles



MISMATCH

Stop and record

Wrong product, short count, or a package that does not belong to this shipment. Evidence is captured on the spot.

escalated with photos attached

No dashboards, no scores, no confidence percentages. A worker on a dock needs a decision, and the reason behind it in one line.

The result becomes the record — nobody types it in again

This is the step that removes the second error opportunity, and the one finance actually cares about.

Goods receipt (GRN)

Line by line, with the attribution behind every unit

Evidence photos

The frames the decision was actually made from

Timestamp + location

When and where the check happened, from the device

Worker and site

Who stood in front of the cargo and signed it off

Discrepancy report

Generated only when there is something to report

Audit trail entry

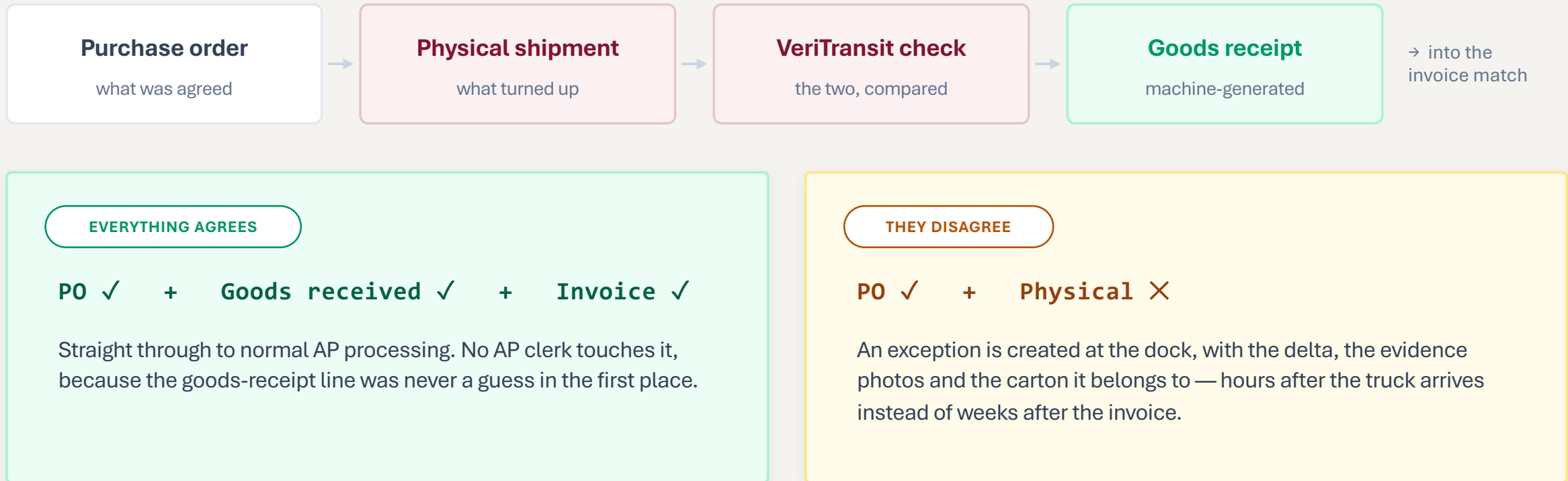
Append-only, so a flagged scan cannot quietly disappear

Offline is the normal case, not the failure case.

Docks, basements and check posts have bad signal. Everything above is produced on the device and queued; when the connection returns it syncs and the server re-verifies it independently. The worker never waits for a network.

We make the physical side of a B2B transaction machine-readable

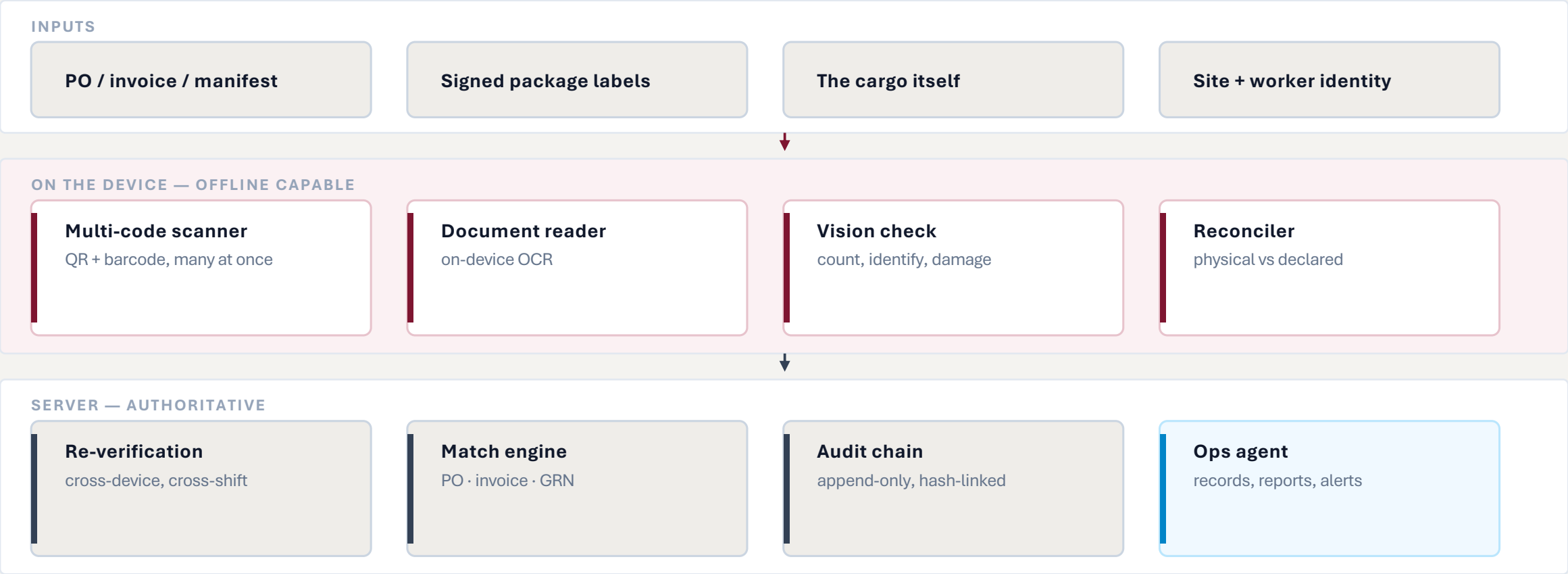
Not a payment system. The missing input to the 3-way match that every AP team already runs.



We are not saying "our AI controls payments." We are saying the physical half of the transaction finally has evidence behind it.

Deterministic checks decide; the model explains and routes

The AI agent sits after verification, never inside it — a misread can cost a supervisor a walk, never a wrong release.



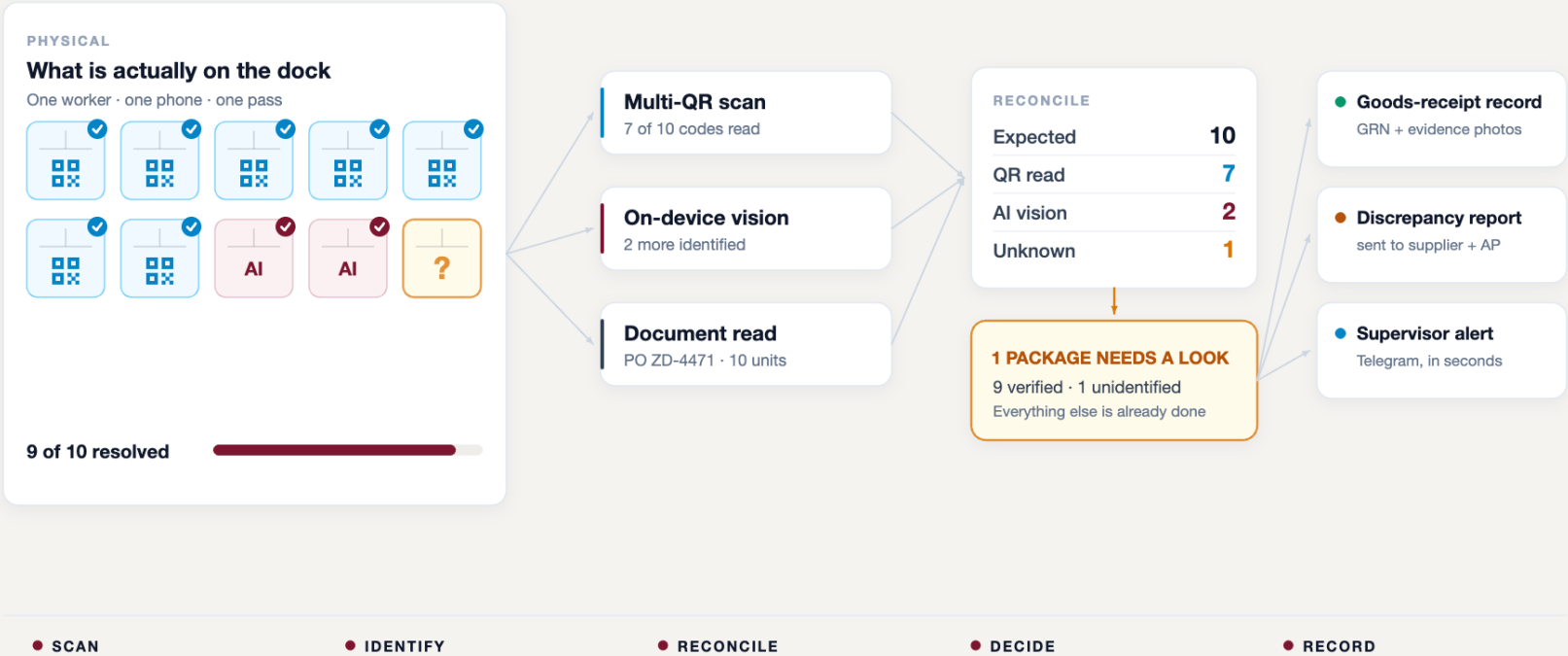
Consumers: ERP / WMS · AP exception queue · Telegram for supervisors · dispute & insurance evidence packs

One pass over ten cartons, from camera to goods receipt

VERITRANSIT · RECEIVING COPILOT

Physical shipment, reconciled against the digital order

Shipment SHP-2026-090231 · PO ZD-4471 · 10 units expected



The worker handled one exception. The record wrote itself.

15-second embedded clip — press play in slideshow. The final frame is the static diagram, so the slide reads correctly in print and PDF.

The agent does not decide whether the cargo matches

It takes a structured, already-decided result and does the chores that follow it.

NOT THE AGENT'S JOB

Judging the match

Whether ten units are present is arithmetic over scans and vision output — rule-based, explainable, reproducible, and testable. A language model has no business in that decision, and putting one there is how a demo becomes an audit finding.

The model can raise a flag. It can never clear one on its own.

WHAT IT ACTUALLY DOES, ONCE THE RESULT EXISTS

Updates the database

shipment, line items, goods-receipt state

Generates the discrepancy report

supplier-ready, with evidence attached

Notifies the right person

supervisor, supplier, AP — by role, not broadcast

Creates the GRN record

in the format the ERP expects

Answers questions in Telegram

"status SHP-90231", "why was it held?"

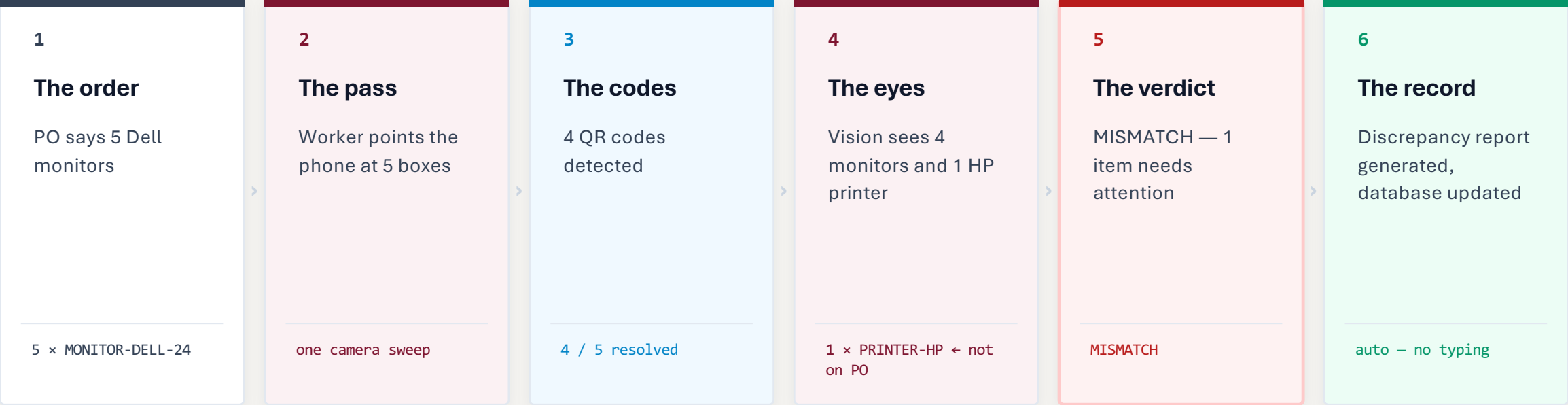
Queues everything offline

and replays it when connectivity returns

Deterministic engines decide. The model explains, drafts and routes — inside a policy it cannot widen.

Thirty seconds, one mismatch, the whole story

Not ten screens. One shipment that is wrong, and what the phone does about it.



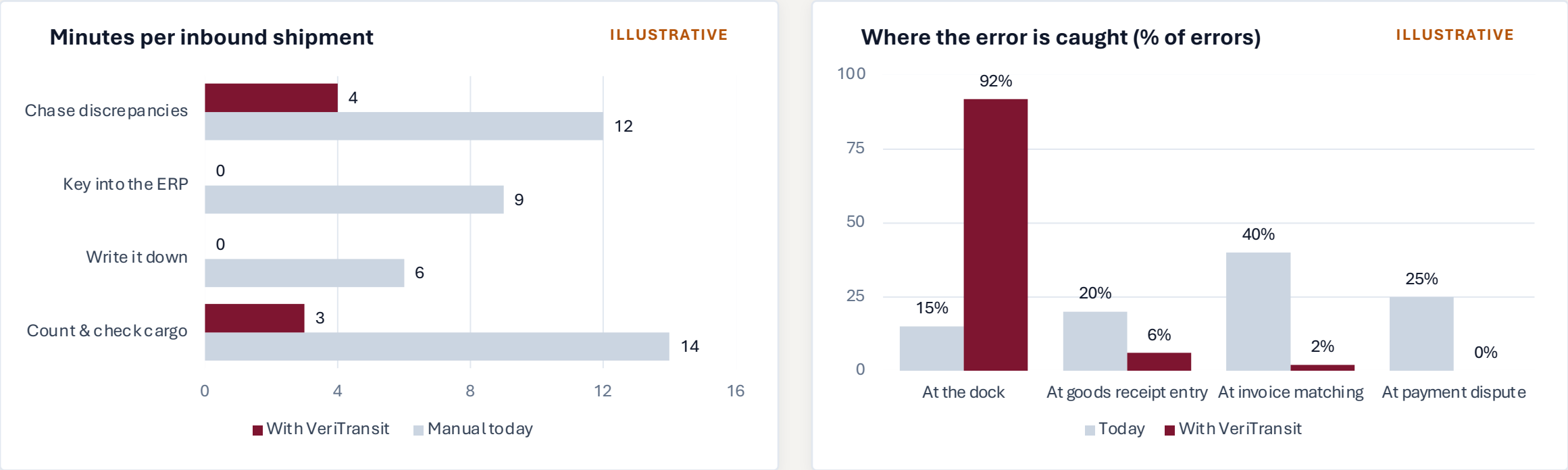
The printer is the point.

A barcode scanner reports four good scans and moves on. Only something that looks at the cargo notices that the fifth box is a different product entirely — and that is exactly the error that surfaces six weeks later as an invoice dispute.

EXPECTED IMPACT

The work shrinks; errors surface while evidence exists

Modelled on a 40-carton inbound. These are targets for the pilot to confirm, not measured results.



Catching a short shipment at the dock is a two-minute conversation. Catching it at invoice matching is a six-week receivable.

Illustrative model, not measured: manual = per-carton visual check, tally sheet and later ERP entry; VeriTransit = one camera pass plus exception handling. Editable in data/charts.json; to be replaced with pilot timings.

The hard parts are built; the pilot is what is left

Kotlin end to end — one set of models shared by the app, the server, the console and the bot.

RUNNING TODAY

- **Android app**
Compose UI, field + warehouse + delivery flows, camera scanning
- **On-device AI**
vision and document reading on the phone's NPU, offline
- **Backend**
Kotlin/Ktor, Postgres, reconciliation and append-only audit chain
- **Admin console**
shipments, documents, discrepancy queue, evidence
- **Telegram agent surface**
query a shipment, receive alerts, approve in chat
- **Signed labels**
Ed25519 tokens verified on the device with no network

NEXT

Pilot on one inbound dock

measure the numbers on slide 17 for real

ERP / WMS write-back

GRN straight into the customer's system

Vendor scorecards

discrepancy history per supplier and route

Payment release signal

verified receipt as a gate, under finance policy

What we need from a pilot partner is one dock, one week, and access to the PO feed.

THE POSITION

AI-powered physical verification for B2B commerce — built as a worker copilot, not another enterprise system.

We do not replace the worker.

One camera pass replaces the counting, the tally sheet and the data entry.

We do not decide the money.

We supply the physical evidence the three-way match has always been missing.

We do not need new hardware.

A phone the site already owns, working offline, on the dock.